Preparation

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1 Day 01: Front-Loading Linear Maps & Matrix Algebra (Basics)

1.1 Block A: Active Computation

1.1.1 Task

1. In this exercise, we work over the field $\mathbb{R}$. Consider $A = \begin{bmatrix} 2 & 1 & -1 \\ 4 & 5 & -3 \\ -2 & 5 & -2 \end{bmatrix} \in \mathbb{R}^{3\times 3}$, $b = \begin{bmatrix} 1 \\ 7 \\ 3 \end{bmatrix} \in \mathbb{R}^3$.

   Perform Gaussian elimination on the system $Ax = b$ without row exchanges. Show explicitly how you eliminate the entries below the diagonal in the first and second columns, and write down the resulting upper-triangular matrix U.

2. Let m_{21}, m_{31} be the multipliers used to eliminate the entries in the first column, and let m_{32} be the multiplier used to eliminate the $(3, 2)$ -entry after the first elimination step. Form the unit lower-triangular matrix

$$L = \begin{bmatrix} 1 & 0 & 0 \\ m_{21} & 1 & 0 \\ m_{31} & m_{32} & 1 \end{bmatrix}$$

   Verify that your matrices satisfy $A = LU$.

3. Solve $Ax = b$ using the factorization $A = LU$ by (i) solving $Ly = b$ for y (forward substitution) and then (ii) solving $Ux = y$ for x (backward substitution). Present your intermediate vector y and the final solution x.

4. (Programming, Julia) Implement two Julia functions `forward_substitution(L,b)` and `backward_substitution(U,y)` that solve $Ly = b$ band $Ux = y$ for triangular matrices of compatible sizes. Use your code to solve the system from this exercise and verify that your result agrees with Julia's $A \setminus b$.

1.1.2 Solution

1. (a)

$$
\begin{aligned}
[A \mid b] &= \left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 4 & 5 & -3 & 7 \\ -2 & 5 & -2 & 3 \end{array}\right] \\
&\xrightarrow{m_{21}=2,\, m_{31}=-1} \left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 0 & 3 & -1 & 5 \\ 0 & 6 & -3 & 4 \end{array}\right] \\
&\xrightarrow{m_{32}=2} \left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 0 & 3 & -1 & 5 \\ 0 & 0 & -1 & -6 \end{array}\right] = [U \mid \tilde{b}]
\end{aligned}
$$

2. (b)

$$
\begin{aligned}
L &= \begin{bmatrix} 1 & 0 & 0 \\ m_{21} & 1 & 0 \\ m_{31} & m_{32} & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}
\end{aligned}
$$

3. (c)

$$
\begin{aligned}
Ux &= y \\
(-1)x_3 &= -6 && \implies x_3 = 6 \\
(3)x_2 + (-1)x_3 &= 5 && \implies x_2 = \frac{11}{3} \\
(2)x_1 + (1)x_2 + (-1)x_3 &= 1 && \implies x_1 = \frac{5}{3}
\end{aligned}
$$

4. (d)

```
function forward_substitution(L, b)
    n = size(L, 1)
    y = zeros(Float64, n)

    for i in 1:n
        sum = b[i]

        for j in 1:(i-1)
            sum -= L[i, j] * y[j]
        end
```

```
        y[i] = sum / L[i, i]
    end
    return y
end

function backward_substitution(U, y)
    n = size(U, 1)
    x = zeros(Float64, n)

    for i in n:-1:1
        sum = y[i]

        for j in (i+1):n
            sum -= U[i, j] * x[j]
        end
        x[i] = sum / U[i, i]
    end
    return x
end

L = [ 1  0  0;
      2  1  0;
     -1  2  1]

U = [ 2  1 -1;
      0  3 -1;
      0  0 -1]

b = [ 1;
      7;
      3]

y = forward_substitution(L, b)
x = backward_substitution(U, y)

println("Intermediate vector y: ", y)
println("Final solution x: ", x)
```

```
forward_substitution (generic function with 1 method)
backward_substitution (generic function with 1 method)
3×3 Matrix{Int64}:
  1  0  0
  2  1  0
 -1  2  1
3×3 Matrix{Int64}:
 2  1  -1
```

```
 0  3  -1
 0  0  -1
3-element Vector{Int64}:
 1
 7
 3
3-element Vector{Float64}:
  1.0
  5.0
 -6.0
3-element Vector{Float64}:
 1.6666666666666667
 3.6666666666666665
 6.0
Intermediate vector y: [1.0, 5.0, -6.0]
Final solution x: [1.6666666666666667, 3.6666666666666665, 6.0]
```

1.1.3 Review Metrics

- Time Spent: 1 hours
- Confidence (1-5): 4
- Mistakes / Takeaways: Mainly focused on best structure approach, balancing speed and error prone.

1.2 Block B: Theory & Proof Memorization

1.2.1 Task

Learn and memorize the Steinitz Exchange Theorem (Theorem III.1.47).

1.2.2 Solution

Let V be a vector space over a field $\mathbb{F}$. Let $L = \{v_1, v_2, \ldots, v_m\}$ be linearly independent and $S = \{w_1, w_2, \ldots, w_n\}$ be a spanning set over V. Then $m \leq n$, and there exists $S' \subseteq S$ with $|S'| = n - m$ such that $S' \cup L$ spans V.

- **Custom Exam Style Proof.**
  - **Theorem.**
    Let $L = \{v_1, \ldots, v_m\}$ be linearly independent and $S = \{w_1, \ldots, w_n\}$ span V. Then $m \leq n$, and $\exists S' \subseteq S$ with $|S'| = n - m$ such that $L \cup S'$ spans V.

- **Proof.**
  $m = 0$: Trivial, $L \cup S = S$ spans V and $0 \leq n$.
  Assume case holds for $m-1$. Applied to $\{v_1, \ldots, v_{m-1}\}$ and S, we have obtain $S_{m-1} \subseteq S$ with $|S_{m-1}| = n - (m-1)$ such that $\{v_1, \ldots, v_{m-1}\} \cup S_{m-1}$ spans V.

Formal Proof.

- **Lemma (Exchange Lemma).**

  Let V be a $\mathbb{F}$ - vector space, $A \subseteq V$ a subset, and $v \in \langle A \rangle$ an arbitrary element.

  `here we just defined that $V$ is our vector space, $A$ is a subset of $V$ (because of notation $\langle A \rangle$), and let $v$ be some vector that lies in the span of $A$`

  Let $a_0 \in A$ be an element such that there is a representation of v as a linear combination of elements of A in which a_0 has a coefficient $\lambda_0 \neq 0$.

  `here we are basically linking $a_0$ to arbitrary element $v$, by saying that $v$ can be represented as a linear combination, where $a_0$ takes place (as $\lambda_0 \neq 0$)`

  Then the set $A' := (A \setminus \{a_0\}) \cup \{v\}$, which is obtained by replacing the vector a_0 in A by v, satisfies $\langle A' \rangle = \langle A \rangle$.

  `here we are constructing a new set $A'$ which is a set $A$ with replaced $v$ over $a_0$, then saying that $\langle A' \rangle = \langle A \rangle$ (meaning they have a same span, which implies $A'$ being a spanning set over $V$)`

- **Lemma Proof (Exchange Lemma).**

  By assumption we have $v \in \langle A \rangle$, hence $A' \subseteq \langle A \rangle$, hence $\langle A' \rangle \subseteq \langle A \rangle$.

  `here as I can see we are starting to prove from both sides (e.g. $\langle A' \rangle \subseteq \langle A \rangle$ and $\langle A \rangle \subseteq \langle A' \rangle$ implies $\langle A \rangle = \langle A' \rangle$). since v was assumed to be a vector that lies in the span of A, A' must be a subset of span A`

  Conversely, by assumption there exists elements $a_1, \ldots, a_n \in A \setminus \{a_0\}$ and scalars $\lambda_0, \ldots, \lambda_n$ with $\lambda_0 \neq 0$ such that $v = \lambda_0 a_0 + \sum_{j=1}^{n} \lambda_j a_j$.

  `here we are just saying that since by assumption $v$ is a linear combination of span $A$, we can write $v$ in the form of $v = \lambda_0 a_0 + \sum_{j=1}^{n} \lambda_j a_j$`

  Hence, $a_0 = \lambda_0^{-1} \left(v - \sum_{j=1}^{n} \lambda_j a_j \right) \in \langle A' \rangle$.

  `here we have just rewrote the previous equiation for $a_0$, since the RHS is fully constructed of $v$ and $a_j$, remembering that $A' := (A \setminus \left\{ a_0 \right\}) \cup \left\{ v \right\}$ we conclude that $a_0 \in \langle A' \rangle$`

  Since obviously $A \setminus \{a_0\} \subseteq \langle A' \rangle$ holds, we obtain $A \subseteq \langle A' \rangle$ and therefore $\langle A \rangle \subseteq \langle A' \rangle$.

  `here we stated obvious statement, and by the work done before (that $a_0 \in \langle A' \rangle$) stated $\langle A \rangle \subseteq \langle A' \rangle$, hence the two sided proof finishes`

- **Theorem (Steinitz Exchange Theorem).**

  Let V be a finitely generated $\mathbb{F}$ vector space, and let $\{v_1, \ldots, v_n\}$ be a spanning set of V. Then:

  1. If $A \subseteq V$ is a linearly independent subset, then A is finite and has at most n elements. If $A = \{a_1, \ldots, a_k\}$ with $k < n$, then one can renumber the elements v_i, in such a way $\{a_1, \ldots, a_k, v_{k+1}, \ldots v_n\}$ is a spanning set of V.
  2. If $A \subseteq V$ is an arbitrary spanning set, then there exists a finite subset $A' \subseteq A$ with at most n elements that is also a spanning set of V.

1.2.3 Review Metrics

- Time Spent:
- Confidence (1-5):
- Mistakes / Takeaways:

1.3 Block C: Application/Testing

1.3.1 Task

1. Define the terms kernel, image, and rank of a linear map. (2 Points)
2. Provide a proper formulation for the Steinitz exchange theorem (Theorem III.1.47). (2 Points)
3. Give a proof of the Steinitz exchange theorem. (6 Points)

1.3.2 Solution

1. (a)

   Let $T : V \to W$ be a linear map between vector spaces V and W over a field $\mathbb{F}$.

   - **Kernel of T:**

     $$ker(T) = \{v \in V : T(v) = 0_W\}$$

     A kernel is the preimage of a zero vector in W. It is a subspace of V.
   - **Image of T:**
     $im(T) = \{T(v) : v \in V\} = T(V)$
     The image is the set of all outputs of T. It is a subspace of W.

- **Rank of T:**

$$rank(T) = dim(im(T))$$

Rank is the dimension of the image space.

2. (b)

Let V be a vector space of a field $\mathbb{F}$. Let $L = \{v_1, v_2, \ldots, v_m\}$ be linearly independent and $S = \{w_1, w_2, \ldots, w_n\}$ be a spanning set over V. Then, $m \leq n$, and there exists $S' \subseteq S$ with $|S'| = n - m$ such that $L \cup S'$ spans V.

3. (c)

1.3.3 Review Metrics

- Time Spent:
- Confidence (1-5):
- Mistakes / Takeaways:

2 Day 02: Matrix Algebra - Inverses & Pivoting

2.1 Block A: Active Computation

2.1.1 Task

Let

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 2 & 2 & 3 \\ 4 & 1 & 8 \end{bmatrix} \in \mathbb{R}^{3\times 3}, \quad b = \begin{bmatrix} 3 \\ 7 \\ 10 \end{bmatrix} \in \mathbb{R}^3.$$

(a) Perform the first step of Gaussian elimination with partial pivoting in the first column, i.e., look for an appropriate row j in your working column that contains a non-zero element, and then swap the first and j th row. Write down the permutation matrix P_1 representing the row swap that is performed (if any), and the resulting matrix $P_1 A$.

(b) Continue the elimination with partial pivoting until you obtain matrices P, L, U such that

$$PA = LU$$

, where P is a permutation matrix, L is unit lower triangular, and U is upper triangular. Write down P, L, and U explicitly.

(c) Solve $Ax = b$ using the factorization $PA = LU$ by solving $Ly = Pb$, $Ux = y$. Present y and r.

2.1.2 Solution

1. (a)

$$
\begin{aligned}
P_1A &= \underbrace{\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}}_{P_1} \underbrace{\begin{bmatrix} 0 & 2 & 1 \\ 2 & 2 & 3 \\ 4 & 1 & 8 \end{bmatrix}}_{A} \\
&= \begin{bmatrix} 4 & 1 & 8 \\ 2 & 2 & 3 \\ 0 & 2 & 1 \end{bmatrix}
\end{aligned}
$$

2. (b)

$$
\begin{aligned}
P_1A &= \begin{bmatrix} 4 & 1 & 8 \\ 2 & 2 & 3 \\ 0 & 2 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 1 & 8 \\ 0 & \frac{3}{2} & -1 \\ 0 & 2 & 1 \end{bmatrix} \qquad \left(R_2 \leftarrow R_2 - (\frac{1}{2})R_1\right) \\
P_2A &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 1 & 8 \\ 0 & \frac{3}{2} & -1 \\ 0 & 2 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ \frac{1}{2} & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 1 & 8 \\ 0 & 2 & 1 \\ 0 & \frac{3}{2} & -1 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{2} & \frac{3}{4} & 1 \end{bmatrix} \begin{bmatrix} 4 & 1 & 8 \\ 0 & 2 & 1 \\ 0 & 0 & -\frac{7}{4} \end{bmatrix} \qquad \left(R_3 \leftarrow R_3 - (\frac{3}{4})R_2\right)
\end{aligned}
$$

$$
\begin{aligned}
P &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \\
L &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{2} & \frac{3}{4} & 1 \end{bmatrix} \\
U &= \begin{bmatrix} 4 & 1 & 8 \\ 0 & 2 & 1 \\ 0 & 0 & -\frac{7}{4} \end{bmatrix}
\end{aligned}
$$

3. (c)

$$Ly = Pb$$
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{2} & \frac{3}{4} & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 3 \\ 7 \end{bmatrix}$$

$$\begin{aligned} (1)y_1 &= 10 & \implies y_1 &= 10 \\ (0)y_1 + (1)y_2 &= 3 & \implies y_2 &= 3 \\ (\frac{1}{2})y_1 + (\frac{3}{4})y_2 + (1)y_3 &= 7 & \implies y_3 &= -\frac{1}{4} \end{aligned}$$

$$Ux = y$$
$$\begin{bmatrix} 4 & 1 & 8 \\ 0 & 2 & 1 \\ 0 & 0 & -\frac{7}{4} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 3 \\ -\frac{1}{4} \end{bmatrix}$$

$$\begin{aligned} (-\frac{7}{4})x_3 &= -\frac{1}{4} & \implies x_3 &= \frac{1}{7} \\ (2)x_2 + (1)x_3 &= 3 & \implies x_2 &= \frac{10}{7} \\ (4)x_1 + (1)x_2 + (8)x_3 &= 10 & \implies x_1 &= \frac{13}{7} \end{aligned}$$

2.1.3 Review Metrics

- Time Spent: 40 minutes, was mainly thinking about the notation of writing.
- Confidence (1-5): 4
- Mistakes / Takeaways: There were some basic arithmetic mistakes that I later fixed.

2.2 Block B: Theory & Proof Memorization

2.2.1 Task

Learn and memorize the Dimension Theorem / Rank-Nullity (Theorem IV.1.10).

2.2.2 Solution

1. Annotated Rewriting

   Theorem (Dimension Theorem for Linear Maps).

   Let V and W be finite-dimensional $\mathbb{F}$ vector spaces, and let $f : V \to W$ be linear. Then

   $$dim(V) = dim(im(f)) + dim(ker(f))$$

   Proof.

   We choose a basis $\{b_1, \ldots, b_k\}$ for the subspace $ker(f) \subseteq V$.

   define basis of a kernel

   Since the vectors b_i are linearly independent, *Corollary III.1.48(ii)* shows they can be extended to a basis $\{b_1, \ldots, b_n\}$ of the vector space V.

   using Corollary to state the basis set can be extended (with linear independent vectors) to span the whole vector space V

   We claim that the vectors $f(b_{k+1}), \ldots, f(b_n) \in W$ form a basis of the subspace $im(f) \in W$, which of course proves the theorem.

   outlining the whole point of the proof, to demonstrate $\{f(b_{k+1}), \ldots, f(b_n)\}$ (extended part of initial kernel basis) is basis of image under f

   First, *Proposition III.2.11* tells us that vectors $f(b_1), \ldots, f(b_n)$ form a generating system of $im(f)$.

   using proposition to state $\{f(b_1), \ldots, f(b_n)\}$ spans the $im(f)$, foundation

   By construction, however, we have $f(b_1) = \ldots = f(b_k) = 0$, so $f(b_{k+1}), \ldots, f(b_n)$ also form a generating system.

   outline that only $\{f(b_{k+1}), \ldots, f(b_n)\}$ actually spans the $im(f)$ (as linearly mapped vectors of kernel result in zero - no contribution)

   On the other hand, assume that $\lambda_{k+1}, \ldots, \lambda_n \in \mathbb{F}$ are scalars such that $\lambda_{k+1} f(b_{k+1}) + \ldots + \lambda_n f(b_n) = 0$ holds.

   assume existence of scalars $\lambda_{k+1}, \ldots, \lambda_n \in \mathbb{F}$ such that, $\lambda_{k+1} f(b_{k+1}) + \ldots + \lambda_n f(b_n) = 0$

   Then we can write the left hand side as $f(\lambda_{k+1} b_{k+1} + \ldots + \lambda_n b_n)$, hence, $\lambda_{k+1} b_{k+1} + \ldots + \lambda_n b_n \in ker(f)$.

   using linearity of f to rewrite the scaled vector sums into the form of $f(\lambda_{k+1} b_{k+1} + \ldots + \lambda_n b_n)$, and hence, as scaled vector sum equals to zero, it is within the kernel of f (by definition of a kernel $ker(T) = \{T(v) = 0 : v \in V\}$)

   By construction, the vectors $b_1, \ldots, b_k$ form a basis of this subspace, so we find scalars $\mu_1, \ldots, \mu_k \in \mathbb{F}$ such that

   $$\lambda_{k+1} b_{k+1} + \ldots + \lambda_n b_n = \mu_1 b_1 + \ldots + \mu_k b_k$$

   /as scaled vector sum (constructed with $b_{k+1}, \ldots, b_n$) is in kernel, it can be written as scaled vector sum of $b_1, \ldots, b_k$, (there must exist scalars such that) $\lambda_{k+1} b_{k+1} + \ldots + \lambda_n b_n = \mu_1 b_1 + \ldots + \mu_k b_k$ /

But this means precisely that

$$\mu_1 b_1 + \ldots + \mu_k b_k - \lambda_{k+1} b_{k+1} - \ldots - \lambda_n b_n = 0$$

rewriting the equation to further invoke property of linear independence of $b_1, \ldots, b_n$

and by linear independence of $b_1, \ldots, b_n$ this is possible only if

$$\mu_1 = \ldots = \mu_k = \lambda_{k+1} = \ldots = \lambda_n = 0$$

as $b_1, \ldots, b_n$ *are linearly independent (as are basis of vector space* V*) the only way the total sum of scaled vectors can be equal to zero iff all scalars equal to zero, hence, as* $f(b_{k+1}), \ldots, f(b_n)$ *are generating system of* $im(f)$ *and also are linear independent, they are a basis of* $im(f)$

to sum up, as $\{b_1, \ldots, b_n\}$ *is basis of vector space, let* $\{b_1, \ldots, b_k\}$ *be basis of* $ker(f)$ *(*$k \leq n$*). the extended part obviously spans* $im(f)$*, and now it only needs to prove its linear independence (by using linearity and scalars operations). as extended part spans* $im(f)$ *and is linear independent, it is a basis of* $im(f)$*, which implies* $dim(V) = dim(ker(f)) + dim(im(f))$

2. Personal Review

   Let V and W be a finite dimensional F vector spaces.

   Let $f : V \to W$ be linear map.

   Let $\{b_1, \ldots, b_k\}$ be a basis of $ker(f) \subseteq V$.

   Proving kernel basis can be extended to n values (keep in mind b_i are all linearly independent), with $\{f(b_{k+1}), \ldots, f(b_n)\}$ being a basis of $im(f) \subseteq W$.

   Claim $\{f(b_1), \ldots, f(b_n)\}$ form a basis of $im(f)$.

   Note $f(b_1) = \ldots = f(b_k) = 0$ (as they are kernel), while only $\{b_{k+1}, \ldots, b_n\}$ are "contributing" to the span of $im(f)$.

   Assume $\exists\, \lambda_{k+1}, \ldots, \lambda_n \in \mathbb{F}$ (scalars) such that $\lambda_{k+1} f(b_{k+1}) + \ldots + \lambda_n f(b_n) = 0$.

   Hence, (rewriting) $f(\lambda_{k+1} b_{k+1} + \ldots + \lambda_n b_n) = 0$.

   Thus, $\lambda_{k+1} b_{k+1} + \ldots + \lambda_n b_n \in ker(f)$ (as their sum is zero).

   Since $\{b_1, \ldots, b_k\}$ are basis, there must exist scalars $\mu_1, \ldots, \mu_k \in \mathbb{F}$ such that $\lambda_{k+1} b_{k+1} + \ldots + \lambda_n b_n = \mu_1 b_1 + \ldots + \mu_k b_k$

   Hence, $\mu_1 b_1 + \ldots + \mu_k b_k - \lambda_{k+1} b_{k+1} - \ldots - \lambda_n b_n = 0$.

   And by linear independence of $b_1, \ldots, b_n$, this is possible iff $\mu_1 = \ldots = \mu_k = \lambda_{k+1} = \ldots = \lambda_n = 0$.

3. Exam Style

   - **Theorem:**
     For a finite dimensional vector space V and a linear map $f : V \to W$, $dim(V) = dim(ker(f)) + dim(im(f))$.
   - **Proof:**
     Let $dim(V) = n$ and $dim(ker(f)) = k$.

(a) Basis Extension
Choose a basis $\{b_1, \ldots, b_k\}$ for $ker(f)$. Extend it to basis $\{b_1, \ldots, b_n\}$ for V.

(b) Claim
The set $S = \{f(b_{k+1}), \ldots, f(b_n)\}$ is basis for $im(f)$.

(c) Proof of Span
Let $y \in im(f)$, then $\exists x \in V : y = f(x)$. Since $\{b_1, \ldots, b_n\}$ is a basis for V, we can write $x = \sum_{i=1}^{n} c_i b_i$. By linearity, and because $b_1, \ldots, b_k \in ker(f)$:

$$y = f\left(\sum_{i=1}^{n} c_i b_i\right) = \sum_{i=1}^{k} c_i f(b_i) + \sum_{i=k+1}^{n} c_i f(b_i) = \sum_{i=k+1}^{n} c_i f(b_i)$$

Thus, the set S spans $im(f)$.

(d) Proof of Linear Independence
Assume $\sum_{i=k+1}^{n} c_i f(b_i) = 0$. By linearity, $f(\sum_{i=k+1}^{n} c_i b_i) = 0$, meaning the vector $v = \sum_{i=k+1}^{n} c_i b_i$ is in $ker(f)$. Hence,
Because $v \in ker(f)$, it can be expressed using the kernel's basis: $v = \sum_{i=1}^{k} c_i b_i$. Hence,

$$\sum_{i=1}^{k} c_i b_i - \sum_{i=k+1}^{n} c_i b_i = 0$$

Because $\{b_1, \ldots, b_n\}$ is a basis of V, all coefficients must be strictly zero. Therefore, $c_{k+1} = \ldots = c_n = 0$, proving S is linearly independent.

(e) Conclusion
Therefore, as S spans $im(f)$ and is linearly independent, it is a basis of $im(f)$. The size of this basis is $n - k$.
Therefore, $dim(V) = dim(ker(f)) + dim(im(f))$.

2.2.3 Review Metrics

- Time Spent: Few hours, learned from zero.
- Confidence (1-5): 5.
- Mistakes / Takeaways: None.

2.3 Block C: Application/Testing

2.3.1 Task

1. Compute all solutions of the following linear system of equations using LU factorization (if needed with partial pivoting). Explicitly provide the factors L, U, and the solution.

$$
\begin{aligned}
6x + 6y + 5z &= 1 \\
2x + 4y + 5z &= 2 \\
4x + 3y + 4z &= 1.
\end{aligned}
$$

2. Compute the inverse matrix of A using elementary row operations

$$
A = \begin{pmatrix} 3 & 2 & 3 \\ 3 & 1 & 3 \\ 2 & 3 & 1 \end{pmatrix}.
$$

(6 Points)

2.3.2 Solution

1. 1.

$$
\begin{aligned}
[A \mid b] &= \left[\begin{array}{ccc|c} 6 & 6 & 5 & 1 \\ 2 & 4 & 5 & 2 \\ 4 & 3 & 4 & 1 \end{array}\right] \\
&\xrightarrow{m_{21}=\frac{1}{3}, m_{31}=\frac{4}{6}} \left[\begin{array}{ccc|c} 6 & 6 & 5 & 1 \\ 0 & 2 & \frac{10}{3} & \frac{5}{3} \\ 0 & -1 & \frac{2}{3} & \frac{2}{6} \end{array}\right] \\
&\xrightarrow{m_{32}=-\frac{1}{2}} \left[\begin{array}{ccc|c} 6 & 6 & 5 & 1 \\ 0 & 2 & \frac{10}{3} & \frac{5}{3} \\ 0 & 0 & \frac{7}{3} & \frac{7}{6} \end{array}\right] = [U \mid y] \\
L &= \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & 1 & 0 \\ \frac{4}{6} & -\frac{1}{2} & 1 \end{bmatrix}
\end{aligned}
$$

$$
\begin{aligned}
(\frac{7}{3})z &= \frac{7}{6} && \implies z = \frac{1}{2} \\
(2)y + (\frac{10}{3})z &= \frac{5}{3} && \implies y = 0 \\
(6)x + (6)y + (5)z &= 1 && \implies x = -\frac{1}{4}
\end{aligned}
$$

2. 2.

$$
\left[\begin{array}{ccc|ccc} 3 & 2 & 3 & 1 & 0 & 0 \\ 3 & 1 & 3 & 0 & 1 & 0 \\ 2 & 3 & 1 & 0 & 0 & 1 \end{array}\right]
$$

$$
\xrightarrow{m_{21}=1, m_{31}=\frac{2}{3}} \left[\begin{array}{ccc|ccc} 3 & 2 & 3 & 1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 1 & 0 \\ 0 & \frac{5}{3} & -1 & -\frac{2}{3} & 0 & 1 \end{array}\right]
$$

$$
\xrightarrow{m_{32}=-\frac{5}{3}} \left[\begin{array}{ccc|ccc} 3 & 2 & 3 & 1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -\frac{7}{3} & \frac{5}{3} & 1 \end{array}\right]
$$

$$
\xrightarrow{m_{13}=-3} \left[\begin{array}{ccc|ccc} 3 & 2 & 0 & -6 & 5 & 3 \\ 0 & -1 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -\frac{7}{3} & \frac{5}{3} & 1 \end{array}\right]
$$

$$
\xrightarrow{m_{12}=-2} \left[\begin{array}{ccc|ccc} 3 & 0 & 0 & -8 & 7 & 3 \\ 0 & -1 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -\frac{7}{3} & \frac{5}{3} & 1 \end{array}\right]
$$

$$
\xrightarrow{R_1 \leftarrow \frac{1}{3}R_1, R_2 \leftarrow -R_2, R_3 \leftarrow -R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{8}{3} & \frac{7}{3} & 1 \\ 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & \frac{7}{3} & -\frac{5}{3} & -1 \end{array}\right]
$$

2.3.3 Review Metrics

- Time Spent: <1 hour
- Confidence (1-5): 3, mainly because I did not practice calculating inverse matrix. Comfortable with LU factorization.
- Mistakes / Takeaways: Only writing style concerns, with arithmetics caution.

3 Day 03: Linear Maps & Systems Integration

3.1 Block A: Active Computation

3.1.1 Task

1. (18) Linear systems of equations, Reduced row echelon form

(a) (12) Solve the following linear system of equations using Gaussian elimination, i.e., an LU-decomposition (possibly with partial pivoting)

$(2+i)z_1 + (-1+3i)z_2 + (4-i)z_3 + (1+2i)z_4 = 5+24i$, $(3-2i)z_1 + (2+i)z_2 + (-1+i)z_3 + 5z_4 = 24-7i$, $(-2+i)z_1 + (1-4i)z_2 + (3+2i)z_3 + (-1+i)z_4 = -14-2i$, $(1+3i)z_1 - 2z_2 + (2-i)z_3 + (4+i)z_4 = 7+10i$.

(b) (6) Transform the following real 4×5 matrix into reduced row echelon form:

$$A = \begin{pmatrix} 12 & 6 & 4 & -7 & -3 \\ -6 & -3 & 12 & 7 & 1 \\ 6 & 3 & -8 & -6 & 4 \\ 18 & 9 & 4 & -11 & 2 \end{pmatrix}$$

3.1.2 Solution

1. (a)

$$\left[\begin{array}{cccc|c} (2+i) & (-1+3i) & (4-i) & (1+2i) & (5+24i) \\ (3-2i) & (2+i) & (-1+i) & (5) & (24-7i) \\ (-2+i) & (1-4i) & (3+2i) & (-1+i) & (-14-2i) \\ (1+3i) & (-2) & (2-i) & (4+i) & (7+10i) \end{array}\right]$$

$$\begin{aligned} |a_{11}| &= |2+i| = \sqrt{5} \\ |a_{21}| &= |3-2i| = \sqrt{13} \\ |a_{31}| &= |-2+i| = \sqrt{5} \\ |a_{41}| &= |1+3i| = \sqrt{10} \end{aligned}$$

$$\begin{aligned} P_1 A &= \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} (2+i) & (-1+3i) & (4-i) & (1+2i) \\ (3-2i) & (2+i) & (-1+i) & (5) \\ (-2+i) & (1-4i) & (3+2i) & (-1+i) \\ (1+3i) & (-2) & (2-i) & (4+i) \end{bmatrix} \\ &= \begin{bmatrix} (3-2i) & (2+i) & (-1+i) & (5) \\ (2+i) & (-1+3i) & (4-i) & (1+2i) \\ (-2+i) & (1-4i) & (3+2i) & (-1+i) \\ (1+3i) & (-2) & (2-i) & (4+i) \end{bmatrix} \end{aligned}$$

$$l_{21} = \frac{2+i}{3-2i} = \frac{(2+i)(3+2i)}{13} = \frac{6+7i+2i^2}{13} = \frac{4+7i}{13}$$

$$
\begin{aligned}
P_1 A &= \begin{bmatrix} (3-2i) & (2+i) & (-1+i) & (5) \\ (2+i) & (-1+3i) & (4-i) & (1+2i) \\ (-2+i) & (1-4i) & (3+2i) & (-1+i) \\ (1+3i) & (-2) & (2-i) & (4+i) \end{bmatrix} \\
\xrightarrow{l_{21}=\frac{4+7i}{13}} & \begin{bmatrix} (3-2i) & (2+i) & (-1+i) & (5) \\ (2+i) & (-1+3i) & (4-i) & (1+2i) \\ (-2+i) & (1-4i) & (3+2i) & (-1+i) \\ (1+3i) & (-2) & (2-i) & (4+i) \end{bmatrix}
\end{aligned}
$$

No, I don't want to solve this bullshit.

3.1.3 Review Metrics

- Time Spent:
- Confidence (1-5):
- Mistakes / Takeaways:

3.2 Block B: Theory & Proof Memorization

3.2.1 Task

Learn and memorize Kernel and Image properties (Proposition III.2.11).

3.2.2 Solution

1. Annotated Rewriting

   Let V and W be vector spaces over $\mathbb{F}$ and let $f : V \to W$ be linear. Then,

   (a) If $\hat{V} \subseteq V$ is a subspace, then $f(\hat{V}) \subseteq W$ is also a subspace. In particular $im(f) \subseteq W$ is a subspace.

   (b) If $\hat{W} \subseteq W$ is a subspace, then $f^{-1}(\hat{W}) \subseteq V$ is also a subspace. In particular, $ker(f) \subseteq V$ is a subspace.

   (c) f is surjective if and only if $im(f) = W$.

   (d) f is injective if and only if $ker(f) = \{0\}$.

   Proofs.

(a) Let $w, w' \in f(\hat{V})$ and $\lambda \in \mathbb{F}$.

define w, w' to be elements of $f(\hat{V})$ (or saying differently, image of subspace $\hat{V}$ under f) and λ (which is element/scalar of a field) to be used for subspace criterion (where given any linear combination $\lambda w + w' \in f(\hat{V})$, given $w, w' \in f(\hat{V})$, $\lambda \in \mathbb{F}$)

Then there exist $v, v' \in \hat{V}$ with $f(v) = w$ and $f(v') = w'$.

letting $w, w' \in f(\hat{V})$ strictly implies existence of $v, v' \in \hat{V} : f(v) = w, f(v') = w'$

By linearity of f we obtain $\lambda w + w' = \lambda f(v) + f(v') = f(\lambda v + v')$.

linearity of f (linear map) implies $\lambda f(v) + f(v') = f(\lambda v + v')$ (remember $\lambda w + w' = \lambda f(v) + f(v')$)

Since $\hat{V}$ is a subspace, we have $\lambda v + v' \in \hat{V}$, and hence $\lambda w + w' \in f(\hat{V})$.

since $\hat{V}$ is a subspace (comes from "if"), $\lambda v + v' \in \hat{V}$ and thus, $\lambda w + w' \in f(\hat{V})$

Since also $f(0) = 0 \in f(\hat{V})$, it follows that $f(\hat{V})$ is a subspace.

as $\hat{V}$ is a subspace $0 \in \hat{V}$ and $f(0) = 0$ (linearity of f) hence $0 \in f(\hat{V})$

(b) Let $v, v' \in f^{-1}(\hat{W})$ and $\lambda \in \mathbb{F}$.

defining tools for subspace criterion, in this case for $f^{-1}(\hat{W})$ (preimage of $\hat{W}$ under f^{-1})

Then $f(v), f(v') \in \hat{W}$, and therefore, $f(\lambda v + v') = \lambda f(v) + f(v') \in \hat{W}$,

letting $v, v' \in f^{-1}(\hat{W})$ strictly implies $f(v), f(v') \in \hat{W}$. linearity of f lets $f(\lambda v + v') = \lambda f(v) + f(v')$ (which are elements of $\hat{W}$). since $\hat{W}$ is a subspace (by "if") it contains any linear combination $\lambda f(v) + f(v')$

since $\hat{W}$ is a subspace, and thus $\lambda v + v' \in f^{-1}(\hat{W})$.

since $\hat{W}$ is a subspace, $f(\lambda v + v') \in \hat{W}$ implies there must exist an element $(\lambda v + v') \in f^{-1}(\hat{W})$ (in a preimage)

We have $0 \in f^{-1}(\hat{W})$ because $f(0) = 0 \in \hat{W}$.

as $f(0) = 0 \in \hat{W}$, following the same logic, $0 \in f^{-1}(\hat{W})$, covering second part of subspace criterion

Therefore $f^{-1}(\hat{W})$ is a subspace.

(c) Is a definition of surjectiveness.

(d) If f is injective, then the preimage of every singleton set has at most one element, so since $f(0) = 0$ it must be that $f^{-1}(\{0\}) = \{0\}$.

proof will be two sided, as proving if and only if

first part of proof, pretty straight forward, if f is injective, then every preimage element is "linked" to unique image element, hence, as $f(0) = 0$, it must be that the preimage of 0 (im image) is $\{0\}$ only

Now assume that $ker(f) = \{0\}$.

second part of proof, assume kernel of f is $\{0\}$ only

Let $x, y \in V$ with $f(x) = f(y)$.

letting $x, y \in V$ with $f(x) = f(y)$ (because we will need to show $x = y$, key idea of injectiveness, no elements of preimage are mapped to the same element of image)

Then we obtain $0 = f(x) - f(y) = f(x - y)$, and hence $x - y \in ker(f)$, therefore $x - y = 0$, so $x = y$.

now we define 0 in image through the difference of $f(x)$ and $f(y)$, and due to linearity of f, $f(x) - f(y) = f(x - y) = 0$, hence, $x - y$ must be element of $ker(f)$, and since kernel consist of only $\{0\}$, x must be equal to y

Thus f is injective.

3.2.3 Review Metrics

- Time Spent: Few hours.
- Confidence (1-5): 4
- Mistakes / Takeaways: Mistakes patched during annotations writing.

3.3 Block C: Application/Testing

3.3.1 Task

1. (6 points) Using elementary row operations, compute the inverse matrix of

$$A = \begin{pmatrix} 3 & 2 & 0 \\ 3 & 1 & 3 \\ 2 & 3 & 1 \end{pmatrix} \in M_3(\mathbb{Z}_5)$$

2. Solve the following:
   (a) (2 points) Define the notions kernel, image, and rank of a linear map.
   (b) (2 points) State the dimension theorem (tn. rank-nullity theorem) for linear maps.
   (c) (6 points) Prove the dimension theorem for linear maps. You do not have to prove additional theorems used in your proof, but you must clearly indicate (refer to) any theorems/propositions you use and state them precisely.

3.3.2 Solution

1. (1)

$$
\begin{aligned}
[A \mid I] &= \left[\begin{array}{ccc|ccc} 3 & 2 & 0 & 1 & 0 & 0 \\ 3 & 1 & 3 & 0 & 1 & 0 \\ 2 & 3 & 1 & 0 & 0 & 1 \end{array}\right] \\
&\xrightarrow{m_{21}=1, m_{31}=\frac{2}{3}} \left[\begin{array}{ccc|ccc} 3 & 2 & 0 & 1 & 0 & 0 \\ 0 & -1 & 3 & -1 & 1 & 0 \\ 0 & \frac{5}{3} & 1 & -\frac{2}{3} & 0 & 1 \end{array}\right] \\
&\xrightarrow{m_{32}=-\frac{5}{3}} \left[\begin{array}{ccc|ccc} 3 & 2 & 0 & 1 & 0 & 0 \\ 0 & -1 & 3 & -1 & 1 & 0 \\ 0 & 0 & 6 & -\frac{7}{3} & \frac{5}{3} & 1 \end{array}\right] \\
&\xrightarrow{m_{23}=3} \left[\begin{array}{ccc|ccc} 3 & 2 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & \frac{1}{6} & \frac{1}{6} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{7}{18} & \frac{5}{18} & \frac{1}{6} \end{array}\right] \\
&\xrightarrow{m_{23}=2} \left[\begin{array}{ccc|ccc} 3 & 0 & 0 & \frac{4}{3} & \frac{1}{3} & -1 \\ 0 & 1 & 0 & -\frac{1}{6} & -\frac{1}{6} & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{7}{18} & \frac{5}{18} & \frac{1}{6} \end{array}\right] \\
&\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{4}{9} & \frac{1}{9} & -\frac{1}{3} \\ 0 & 1 & 0 & -\frac{1}{6} & -\frac{1}{6} & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{7}{18} & \frac{5}{18} & \frac{1}{6} \end{array}\right] = \left[I \mid A^{-1}\right]
\end{aligned}
$$

2. (2)

- (a)
  Let $T : V \to W$ be a linear map between vector spaces V and W over a field $\mathbb{F}$.
  - **Kernel**
    $$ker(T) = \{v \in V : T(v) = 0_W\}$$
  - **Image**
    $$im(T) = \{T(v) : v \in V\} = T(V)$$
  - **Rank**
    $$rank(T) = dim(im(T))$$
- (b)
  Let V be finite dimensional vector space and $f : V \to W$ be linear. Hence,
  $$dim(V) = dim(ker(f)) + dim(im(f))$$
- (c)
  Let $dim(V) = n$, $dim(ker(f)) = k$.
  (a) Basis extension
  Choose basis $\{b_1, \ldots, b_k\}$ for $ker(f)$. Extend basis $\{b_1, \ldots, b_n\}$ for V.
  (b) Claim
  The set $S = \{f(b_{k+1}), \ldots, f(k_n)\}$ is basis for $im(f)$.

(c) Proof of span
Let $y \in im(f)$. Hence, $\exists x \in V : y = f(x)$. Since $\{b_1, \ldots, b_n\}$ is basis for V, we can write $x = \sum_{i=1}^{n} c_i b_i$. By linearity, and because $b_1, \ldots, b_k \in ker(f)$:

$$y = f(\sum_{i=1}^{n} c_i b_i) = \sum_{i=1}^{k} c_i f(b_i) + \sum_{i=k+1}^{n} c_i f(b_i) = \sum_{i=k+1}^{n} c_i f(b_i)$$

Thus, S spans the $im(f)$.

(d) Proof of linear independence
Assume $\sum_{i=k+1}^{n} c_i f(b_i) = 0$. By linearity $f(\sum_{i=k+1}^{n} c_i b_i) = 0$, meaning vector $v = \sum_{i=k+1}^{n} c_i + b_i$ is in $ker(f)$. Hence,
Because $v \in ker(f)$, it can be expressed using the kernel's basis: $v = \sum_{i=1}^{k} c_i b_i$. Hence,

$$\sum_{i=1}^{k} c_i b_i - \sum_{i=k+1}^{n} = 0$$

Because $\{b_1, \ldots, b_k\}$ is basis of V, all coefficients must be strictly zero. Therefore, $c_{k+1} = \ldots = c_n = 0$, proving S is linearly independent.

(e) Conslusion
Therefore, as S spans $im(f)$ and is linearly independent, it is a basis of $im(f)$. Therefore,

$$dim(V) = dim(ker(f)) + dim(im(f))$$

3.3.3 Review Metrics

- Time Spent:
- Confidence (1-5):
- Mistakes / Takeaways:

4 Day 04: Matrix Algebra Mastery & Vector Space Intro

4.1 Block A: Active Computation

4.1.1 Task

Consider

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix} \in \mathbb{R}^{3\times 3}.$$

(a) Perform Gaussian elimination with partial pivoting and determine the rank of A from the resulting echelon form.

(b) For the two right-hand sides

$$b^{(1)} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \quad b^{(2)} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

decide for each system $Ax = b^{(j)}$ whether it is consistent, i.e., whether it has a solution. If consistent, describe the full solution set.

4.1.2 Solution

1. (a)

$$\begin{aligned} P_1 A &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix} \end{aligned}$$

$$\xrightarrow{m_{21}=\frac{1}{2}, m_{31}=\frac{1}{2}} \begin{bmatrix} 2 & 4 & 6 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix}$$

$$\begin{aligned} P_2(P_1 A) &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 4 & 6 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 4 & 6 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{2} & 0 & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 2 & 4 & 6 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

The matrix U has a 2 non zero rows, hence, $Rank(A) = 2$.

2. (b)

- $b^{(1)}$

$$\left[L \mid \tilde{b}\right] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ \frac{1}{2} & 1 & 0 & 1 \\ \frac{1}{2} & 0 & 1 & 1 \end{array}\right]$$

$$\begin{aligned} (1)y_1 &= 2 && \implies y_1 = 2 \\ (\frac{1}{2})y_1 + (1)y_2 &= 1 && \implies y_2 = 0 \\ (\frac{1}{2})y_1 + (0)y_2 + (1)y_3 &= 1 && \implies y_3 = 0 \end{aligned}$$

$$[U \mid y] = \left[\begin{array}{ccc|c} 2 & 4 & 6 & 2 \\ 0 & -1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\begin{aligned} (0)x_3 &= 0 && \implies x_3 = t : t \in \mathbb{R} \\ (-1)x_2 + (-2)x_3 &= 0 && \implies x_2 = -2t \\ (2)x_1 + (4)x_2 + (6)x_3 &= 2 && \implies x_1 = 1 + t \end{aligned}$$

$$x = \begin{bmatrix} 1+t \\ -2t \\ t \end{bmatrix} : t \in \mathbb{R}$$

- $b^{(2)}$

$$\left[L \mid \tilde{b}\right] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ \frac{1}{2} & 1 & 0 & 1 \\ \frac{1}{2} & 0 & 1 & 1 \end{array}\right]$$

$$
\begin{aligned}
(1)y_1 &= 3 && \implies y_1 = 3 \\
(\frac{1}{2})y_1 + (1)y_2 &= 1 && \implies y_2 = -\frac{1}{2} \\
(\frac{1}{2})y_1 + (0)y_2 + (1)y_3 &= 1 && \implies y_3 = -\frac{1}{2}
\end{aligned}
$$

$$
[U \mid y] = \left[\begin{array}{ccc|c} 2 & 4 & 6 & 3 \\ 0 & -1 & -2 & -\frac{1}{2} \\ 0 & 0 & 0 & -\frac{1}{2} \end{array}\right]
$$

Since $(0)x_3 \neq -\frac{1}{2}$, the system is inconsistent and thus have no solution.

4.1.3 Review Metrics

- Time Spent:
- Confidence (1-5):
- Mistakes / Takeaways:

4.2 Block B: Theory & Proof Memorization

4.2.1 Task

Learn and memorize the Dimension Theorem for Sums (Proposition IV.1.4).

4.2.2 Solution

1. Annotated Rewriting

   Proposition (Sum of subspaces)

   Let W and $\hat{W}$ be subspaces of a $\mathbb{F}$ vector space V. Then:

   (a) An element $v \in V$ lies in $W + \hat{W}$ iff there exists elements $w \in W$ and $\hat{w} \in \hat{W} : v = w + \hat{w}$.

   (b) If W and $\hat{W}$ are finite dimensional, then $W + \hat{W}$ is also finite dimensional and

   $$dim(W + \hat{W}) = dim(W) + dim(\hat{W}) - dim(W \cap \hat{W})$$

   Proof.

- Setup
  Let $U, W \in V$ be finite dimensional subspaces.
  Let $I := U \cap W$.
  Choose B_I be basis of I.
- Extend Basis
  Extend B_I to be basis of U:

$$B_U = B_I \cup B_1$$

  Extend B_I to be basis of W:

$$B_W = B_I \cup B_2$$

  Observe:

$$dim(U) = |B_U| = |B_I| + |B_1|$$

$$dim(W) = |B_W| = |B_I| + |B_2|$$

- Claim 1: Span
  Claim, $B_I \cup B_1 \cup B_2$ spans $U + W$.
  Let $v \in U+W$, hence, $v = u+w : u \in U, w \in W$. u can be written as a linear combination of basis $B_U = B_I + B_1$, and w as a linear combination of $B_W = B_I + B_2$, hence, $u + w$ can be represented as a linear combination of $B_I \cup B_1 \cup B_2$. Hence, $B_I \cup B_1 \cup B_2$ spans $U + W$.
- Claim 2: Linear Independence
  Claim, $B_I \cup B_1 \cup B_2$ is linearly independent.
  Assume

$$\sum_{b \in B_I} \alpha_b b + \sum_{b \in B_1} \beta_b b + \sum_{b \in B_2} \gamma_b b = 0$$

  Observe that

$$\left(\sum_{b \in B_I} \alpha_b b + \sum_{b \in B_1} \beta_b b \right) + \left(\sum_{b \in b_2} \gamma_b b \right) = 0$$

  where first parenthesis is a vector in U (as constructed with B_I and B_1), while second paranthesis is a vector in W (as constructed with B_2).

$$\left(\sum_{b \in B_I} \alpha_b b + \sum_{b \in B_1} \beta_b b \right) = - \left(\sum_{b \in b_2} \gamma_b b \right)$$

  Hence, all scalars β must be equal to zero.
  As β's equal to zero, the original equality simplifies to:

$$\sum_{b \in B_I} \alpha_b b + \sum_{b \in B_2} \gamma_b b = 0$$

Where the left side is a linear combination of W basis. This forces all γ scalars to be equal to zero, due to linear independence of the basis.

Hence, all coefficients are zero. Thus, the original vector combination of basis B_I, B_1, B_2 is linear independent.

- Conclusion

  As $B_I \cup B_1 \cup B_2$ spans $U + W$, and is linearly independent, it is a basis of $U + W$. Therefore,

$$dim(U + W) = dim(U) + dim(W) - dim(U \cap W)$$

2. Exam Style

   Proof.

   Let U_1, U_2 be finite dimensional subspaces of V. Let $\{w_1, \cdots, w_k\}$ be basis of $U_1 \cap U_2$. Extend to a basis $\{w_1, \cdots, w_k, u_1 \cdots, u_p\}$ of U_1 and a basis $\{w_1, \cdots, w_k, v_1, \cdots, v_q\}$ of U_2.

   Claim: $B = \{w_1, \cdots, w_k, u_1, \cdots, u_p, v_1, \cdots, v_q\}$ is basis of $U_1 + U_2$.

   Spanning is immediate: any $x = x_1 + x_2 \in U_1 + U_2$ can be expanded to respective basis, giving a linear combination over B.

   Linear independence: Suppose $\sum \alpha_i w_i + \sum \beta_j u_j + \sum \gamma_l v_l = 0$. Then,

$$\sum \gamma_l v_l = -\sum \alpha_i w_i - \sum \beta_j u_j \in U_1$$

   The left side is in U_2, so its vector is in $U_1 \cap U_2$, hence expressible as $\sum \mu_i w_i$. Then $\sum \gamma_l v_l - \sum \mu_i w_i = 0$, and the linear independence of U_2 basis gives $\gamma_l = 0$ for all l. Substituting back, $\sum \alpha_i w_i + \sum \beta_i u_i = 0$, and the linear independence of U_1 basis gives $\alpha_i = \beta_j = 0$ for all i, j.

   Thus, $|B| = k + p + q = m + n - k$, i.e. $dim(U_1 + U_2) = dim(U_1) + dim(U_2) - dim(U_1 \cap U_2)$.

3. Exam Style 2

   Let U_1, U_2 be finite dimensional subspaces of V. Put $I := U_1 \cap U_2$ and choose a basis $B_I = \{w_1, \cdots, w_k\}$ of I. Extend B_I to basis B_1 and B_2:

$$B_{U_1} = \{w_1, \cdots, w_k, u_1, \cdots, u_p\}, \quad B_{U_2} = \{w_1, \cdots, w_k, v_1, \cdots, v_q\}$$

   Consider $B = B_I \cup \{u_1, \cdots, u_p\} \cup \{v_1, \cdots, v_q\}$.

   Every $x \in U_1 + U_2$ equals to $x_1 + x_2 : x_1 \in U_1, x_2 \in U_2$, so B spans U_1, U_2.

   For independence, suppose $\sum \alpha_i w_i + \sum \beta_j u_j + \sum \gamma_l v_l = 0$. Rearanging gives:

$$\sum \gamma_l v_l = -\left\{\sum \alpha_i w_i + \sum \beta_j u_j\right\} \in U_1$$

   but the left side is in U_2, so $\sum \gamma_l v_l \in U_1 \cap U_2 = I$. Hence, $\sum \gamma_l v_l = \sum \mu_i w_i$ for some scalars μ_i. Thus $\sum \gamma_l v_l - \sum \mu_i w_i = 0$, and linear independence of B_{U_2} forces $\gamma_l = \mu_i = 0$. Substituting

back, $\sum \alpha_i w_i + \sum \beta_j u_j = 0$, and linear independence of B_{U_1} forces $w_i = u_j = 0$. Therefore B is a basis of $U_1 + U_2$.

Thus,

$$dim(U_1 + U_2) = k + p + q = (p + k) + (q + k) - k = |U_1| + |U_2| - |U_1 \cap U_2|$$

4. Exam Style 3

   Let U_1, U_2 be finite dimensional subspaces of V. Put $I := U_1 \cap U_2$ and choose a basis $B_I = \{w_1, \cdots, w_k\}$ of I. Extend B_I to basis B_{U_1} B_{U_2}:

$$B_{U_1} = \{w_1, \cdots, w_k, u_1, \cdots, u_p\}, \quad B_{U_2} = \{w_1, \cdots, w_k, v_1, \cdots, v_q\}$$

   Consider $B = B_I \cup \{u_1, \cdots, u_p\} \cup \{v_1, \cdots, v_q\}$.

   Every $x \in U_1 + U_2$ equals to $x_1 + x_2 : x_1 \in U_1, x_2 \in U_2$, so B spans $U_1 + U_2$.

   For independence, suppose $\sum \alpha_i w_i + \sum \beta_j u_j + \sum \gamma_l v_l = 0$. Rearanging gives:

$$\sum \gamma_l v_l = -\left(\sum \alpha_i w_i + \sum \beta_j u_j\right) \in U_1$$

   but the left side is in U_2, so $\sum \gamma_l v_l \in U_1 \cap U_2 = I$. Hence, $\sum \gamma_l v_l = \sum \mu_i w_i$ for some scalars μ_i. Thus, $\sum \gamma_l v_l - \sum \mu_i w_i = 0$, and linear independence of B_{U_1}, forces $\gamma_l = \mu_i = 0$. Therefore, B is a basis of $U_1 + U_2$.

   Thus,

$$dim(U_1 + U_2) = k + p + q = (p + k) + (q + k) - k = |U_1| + |U_2| - |U_1 \cap U_2|$$

5. Exam Style From Memory

   Let U_1, U_2 be finite dimensional subspaces of vector space V. Define $I := U_1 \cap U_2$ and choose basis $B_I = \{w_1, \cdots, w_k\}$. Extend B_I to basis U_1 and U_2:

$$B_{U_1} = \{w_1, \cdots, w_k, u_1, \cdots, u_p\} \quad B_{U_2} = \{w_1, \cdots, w_k, v_1, \cdots, v_q\}$$

   Claim: constructed $B = \{w_1, \cdots, w_k, u_1, \cdots, u_p, v_1, \cdots v_q\}$ is basis of $U_1 + U_2$.

   Span is immediate. Let $x \in U_1 + U_2$. It can be represented as $x = x_1 + x_2 : x_1 \in U_1, x_2 \in U_2$, hence it can be represented as scalar sum of their basis B_{U_1} and B_{U_2}.

   Linear independence. Assume $\sum \alpha_i w_i + \sum \beta_j u_j + \sum \gamma_l v_l = 0$. Rearange,

$$\sum \gamma_l v_l = -\left(\sum \alpha_i w_i + \sum \beta_j u_j\right) \in U_1$$

   As LHS is in U_2, the RHS must be in $U_1 \cap U_2$, and hence could be written as $\sum \mu_i w_i$. Thus,

$$\sum \gamma_l v_l - \sum \mu_i w_i = 0$$

   Linear independence of B_{U_2} forces all $\gamma_l = 0$. Substituting back:

$$\sum \alpha_i w_i + \sum \beta_j u_j = 0$$

Linear independence of B_{U_1} forces $\alpha_i = \beta_j = 0$. Hence B is linearly independent. Therefore, as B is basis of $U_1 + U_2$,

$$dim(U_1 + U_2) = (p + k) + (q + k) - k = dim(U_1) + dim(U_2) - dim(U_1 \cap U_2)$$

4.2.3 Review Metrics

- Time Spent: alot
- Confidence (1-5): 5 (extremely)
- Mistakes / Takeaways: none

4.3 Block C: Application/Testing

4.3.1 Task

(d) Compute the kernel of the linear map $T : \mathbb{C}^2 \to \mathbb{C}^2$

$$T(z, w) := \begin{pmatrix} (1 + 5i)z + 2w \\ (8 + i)z + (1 - 3i)w \end{pmatrix}$$

Is the map injective? (3 Points)

4.3.2 Solution

$$A = \begin{bmatrix} (1 + 5i) & (2) \\ (8 + i) & (1 - 3i) \end{bmatrix}$$
$$A \begin{bmatrix} z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} det(A) &= (1 + 5i)(1 - 3i) - (2)(8 + i) \\ &= 16 + 2i - 16 - 2i \\ &= 0 \end{aligned}$$

As the $det(A) = 0$, this implies matrix is not invertable. Therefore,

- T is not injective
- $ker(T) \neq 0$

$$\begin{bmatrix} (1+5i) & (2) \\ (8+i) & (1-3i) \end{bmatrix} \begin{bmatrix} z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\frac{(8+i)}{(1+5i)} = \frac{(8+i)(1-5i)}{1^2 - 25i^2} = \frac{13-39i}{26} = \frac{1-3i}{2}$$

$$(1-3i) - (\frac{1-3i}{2})(2) = 0$$

Hence the second row becomes 0.

So the system reduces to:

$$(1+5i)z + 2w = 0$$
$$w = -\frac{1+5i}{2}z$$

Thus,

$$ker(T) = \left\{ z \begin{pmatrix} 1 \\ -\frac{1+5i}{2} \end{pmatrix} : z \in \mathbb{C} \right\}$$

4.3.3 Review Metrics

- Time Spent: 40 minutes (as I was learning from 0).
- Confidence (1-5): 4
- Mistakes / Takeaways: Plenty, as I was learning.

5 Day 05: Vector Spaces - Subspaces & Bases

5.1 Block A: Active Computation

5.1.1 Task

1. (12) Subspace

We consider the vector space $\mathbb{F}^n$ for the specified field and some $4 \leq n \in \mathbb{N}$. Check, whether the following subsets are subspaces.

(a) (2) $W_1 := \{x \in \mathbb{R}^n | x_1 + x_2 = 0\}$,

(b) (2) $W_2 := \{x \in \mathbb{R}^n | \sum_{j=1}^{n} j x_j = 0\}$,

(c) (2) $W_3 := \{x \in \mathbb{R}^n | x_1 - x_3 = x_4 + 1\}$,

(d) (2) $W_4 := \{x \in \mathbb{R}^n | \sum_{i=1}^{n} x_i^2 = 0\}$,

(e) (2) $W_5 := \{x \in \mathbb{F}_2^3 | x_1^2 + x_2 + x_3^3 = 0\}$,

(f) (2) $W_6 := \{x \in \mathbb{C}^2 | x_1 \neq 0 \wedge \frac{x_2}{x_1} = 1\}$.

5.1.2 Solution

To prove a set is a subspace I must check three conditions:

1. $0 \in W$
2. Closure under addition ($u, v \in W \implies u + v \in W$)
3. Closure under scalar multiplication ($\lambda \in \mathbb{F}, u \in W \implies \lambda u \in W$)

1. (a)

   (a) Zero Vector
   For $x = 0$, $x_1 + x_2 = 0 + 0$. Thus, $0 \in W_1$.

   (b) Closure Under Addition
   Let $x, y \in W_1$. $(x_1 + y_1) + (x_2 + y_2) = (x_1 + x_2) + (y_1 + y_2) = 0 + 0 = 0 \implies x + y \in W_1$

   (c) Closure Under Scalar Multiplication
   Let $x \in W_1$ and $\lambda \in \mathbb{R}$.
   $(\lambda x_1) + (\lambda x_2) = \lambda(x_1 + x_2) = \lambda(0) = 0 \implies \lambda x \in W_1$

   Hence, W_1 is a subspace.

2. (b)

   (a) Zero Vector
   $\sum j(0) = 0$. Thus, $0 \in W_2$

(b) Closure Under Addition
Let $x, y \in W_2$.
$\sum_{j=1}^{n} j(x_j + y_j) = \sum_{j=1}^{n} jx_j + \sum_{j=1}^{n} jy_j = 0 + 0 = 0$

(c) Closure Under Multiplication
Let $x \in W_2$ and $\lambda \in \mathbb{R}$.
$\sum_{j=1}^{n} j(\lambda x_j) = \lambda \sum_{j=1}^{n} jx_j = \lambda(0) = 0$

Hence, W_2 is a subspace.

3. (c)

(a) Zero Vector
For $x = 0$, $x_1 - x_3 - x_4 = 1 \implies 0 = 1$.
The zero is not in W_3.

Hence, W_3 is not a subspace.

4. (d)

(a) Zero Vector
For $x = 0$, $\sum_{i=1}^{n} 0^2 = 0$. Hence $\{0\} = W_4$.

(b) Closure Under Addition
W_4 is a trivial subspace. It satisfies all axioms $(0 + 0 = 0, \lambda 0 = 0)$.

Hence W_4 is a subspace.

5. (e)

- Zero Vector
$\left\{x \in \mathbb{F}_2^3 | x_1^2 + x_2 + x_3^3 = 0\right\} = \left\{x \in \mathbb{F}_2^3 | x_1 + x_2 + x_3 = 0\right\}$
For $x = 0$, $x_1 + x_2 + x_3 = 0 + 0 + 0 = 0$. Thus $0 \in W_5$.
- Closure Under Addition
Let $x, y \in W_5$.
$(x_1 + y_1) + (x_2 + y_2) + (x_3 + y_3) = (x_1 + x_2 + x_3) + (y_1 + y_2 + y_3) = 0$
- Closure Under Multiplication
Let $x \in W_5$ and $\lambda \in \mathbb{F}_2$.
$(\lambda x_1) + (\lambda x_2) + (\lambda x_3) = \lambda(x_1 + x_2 + x_3) = \lambda(0) = 0$

Hence, W_5 is a subspace.

6. (f)

- Zero Vector
The definition requires $x_1 \neq 0$. The zero vector $0 = (0, 0)$ has the first component 0.

Hence W_6 is not a subspace.

5.1.3 Review Metrics

- Time Spent: 1 hour (new topic, new exercise).
- Confidence (1-5): 4, understood, but might struggle with new style problems.
- Mistakes / Takeaways:

5.2 Block B: Theory & Proof Memorization

5.2.1 Task

5.2.2 Solution

5.2.3 Review Metrics

- Time Spent:
- Confidence (1-5):
- Mistakes / Takeaways:

5.3 Block C: Application/Testing

5.3.1 Task

1. Show that the set $\{p \in \mathbb{R}[X] \mid p(0) = p(1) = 0\}$ is a subspace of the space of polynomials. (3 Points)
2. Is the following set a subspace of the vector space $\mathbb{R}^3$?

$$W := \{x \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 0\}?$$

   Provide an explanation for your answer. (1 Point)

5.3.2 Solution

1. (1)
   $V = \{p \in \mathbb{R}[X] \mid p(0) = p(1) = 0\}$
   - Zero Vector
     Let z be zero polynomial.
     Hence, $z(0) = z(1) = 0$.

- Closure Under Addition
  Let $p, q \in V$.
  $(p+q)(0) = p(0) + q(0) = 0$
  $(p+q)(1) = p(1) + q(1) = 0$
  Thus, $(p+q) \in V$.
- Closure Under Multiplication
  Let $p \in V$ and $\lambda \in \mathbb{R}$.
  $(\lambda p)(0) = \lambda \cdot p(0) = \lambda \cdot 0 = 0$
  $(\lambda p)(1) = \lambda \cdot p(1) = \lambda \cdot 0 = 0$
  Thus, $(\lambda p) \in V$.

Hence, V is subspace.

2. (2)

   - Zero Vector
     Since $x^2+y^2+z^2$ can not be negative by definition, the only possible way $x^2+y^2+z^2 = 0$ iff $x = y = z = 0$.
     Thus, $0 \in W$.
   - Scalability
     Since the only element subspace is $(0,0,0) \in W$ (making subspace trivial), it satisfies all subspace constraints.

   Hence, W is subspace.

5.3.3 Review Metrics

- Time Spent: 30 minutes
- Confidence (1-5): 4
- Mistakes / Takeaways: no

6 Day 06: Linear Systems & Vector Spaces (Independence)

6.1 Block A: Active Computation

6.1.1 Task

Spanning sets, Linear Independence

1. (2) Show that $\mathbb{C}$ is a vector space over $\mathbb{R}$ and find a small spanning set for $\mathbb{C}$.
2. (4) We consider the vector space $\mathbb{F}^4$ for some field $\mathbb{F}$. Show that the vectors $(1,0,0,0)^T$, $(1,1,0,0)^T$, $(1,1,1,0)^T$ and $(1,1,1,1)^T$ are linearly independent.

3. (6) Show that the three vectors $(i, 1, 2+i)$, $(1, 3i+1, 2)$ and $(-1, 5i, 5)$ are a spanning set of $\mathbb{C}^3$. Are they linearly independent?

6.1.2 Solution

1. (1)

   Definition (Vector Space) A vector space V over a field $\mathbb{F}$ is equipped with two operations:

   - Vector addition $+ : V + V \to V$
   - Vector multiplication $\times : \mathbb{F} \times V \to V$

   Such that the following axioms hold for all $u, v, w \in V$ and all scalars $a, b \in \mathbb{F}$:

   (a) Associativity of addition
   $(u + v) + w = u + (v + w)$

   (b) Commutativity of addition
   $u + v = v + u$

   (c) Additive identity
   $\exists 0 \in V : \forall u \in V, u + 0 = u$

   (d) Additive inverse
   $\forall u \in V, \exists -u \in V : u + (-u) = 0$

   (e) Distributivity over scalar additon
   $(a + b) \times u = (a \times u) + (b \times u)$

   (f) Distributivity over vector addition
   $a \times (u + v) = (a \times u) + (a \times v)$

   (g) Compatability of scalar multiplication
   $a \times (b \times u) = (a \times b) \times u$

   (h) Scalar multiplicative identity
   $1 \times u = u$
   Where 1 is a multiplicative identity of $\mathbb{F}$.

   We consider:

   - Vector set: $V = \mathbb{C}$
   - Field of scalars: $\mathbb{F} = \mathbb{R}$
   - Vector addition: usual complex addition
   - Scalar multiplication: multiplication of a real with complex number

   Additive structure:

   (a) Addition is associative

   (b) Addition is commutitative

   (c) The additive identity 0 exists

   (d) Every $z \in \mathbb{C}$ has additive inverse $-z$

   Scalar multiplication sturcture: Let $w, z \in \mathbb{C}$ and $a, b \in \mathbb{R}$

(a) $w(a+b) = wa + wb$
(b) $a(w+z) = aw + az$
(c) $a(bw) = (ab)w$
(d) $1z = z$

Thus, all vector space axioms are satisfied.

Hence, $\mathbb{C}$ is a vector space over $\mathbb{R}$.

Every complex number can be represented as:

$$z = a + bi : a, b \in \mathbb{R}$$

Hence, $z = a \cdot 1 + b \cdot i$.

Therefore, $\mathbb{C} = span_{\mathbb{R}} \{1, i\}$

2. (2)

To show linear independence, we consider the linear system:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} (1) \cdot c_4 &= 0 & \implies c_4 &= 0 \\ (1) \cdot c_3 + (1) \cdot c_4 &= 0 & \implies c_3 &= 0 \\ (1) \cdot c_2 + (1) \cdot c_3 + (1) \cdot c_4 &= 0 & \implies c_2 &= 0 \\ (1) \cdot c_1 + (1) \cdot c_2 + (1) \cdot c_3 + (1) \cdot c_4 &= 0 & \implies c_1 &= 0 \end{aligned}$$

Hence, the only solution $c = 0$, hence, the vectors are linearly independent.

3. (3)

Let v_1, v_2, v_3 be the columns of the matrix A:

$$A = \begin{bmatrix} i & 1 & -1 \\ 1 & 3i+1 & 5i \\ 2+i & 2 & 5 \end{bmatrix}$$

- Determine Rank

$$\begin{aligned} P_1 A &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i & 1 & -1 \\ 1 & 3i+1 & 5i \\ 2+i & 2 & 5 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 3i+1 & 5i \\ i & 1 & -1 \\ 2+i & 2 & 5 \end{bmatrix} \end{aligned}$$

$$
\begin{aligned}
m_{21} &= i \\
(2,2) &= 1 - (i)(3i+1) = 1 + 3 - i = 4 - i \\
(2,3) &= -1 - (i)(5i) = -1 + 5 = 4 \\
&\implies \begin{bmatrix} 1 & 3i+1 & 5i \\ 0 & 4-i & 4 \\ 2+i & 2 & 5 \end{bmatrix}
\end{aligned}
$$

$$
\begin{aligned}
m_{31} &= 2 + i \\
(3,2) &= 2 - (2+i)(3i+1) = 2 - (6i + 2 - 3 + i) = 3 - 7i \\
(3,3) &= 5 - (2+i)(5i) = 5 - (10i - 5) = 10 - 10i \\
&\implies \begin{bmatrix} 1 & 3i+1 & 5i \\ 0 & 4-i & 4 \\ 0 & 3-7i & 10-10i \end{bmatrix}
\end{aligned}
$$

$$
\begin{aligned}
m_{32} &= \frac{(3-7i)(4+i)}{(4^2 - i^2)} = \frac{19 - 25i}{17} \\
(3,3) &= 10 - 10i - (\frac{19-25i}{17})(4) = \frac{170 - 170i - 76 + 100i}{17} = \frac{94 - 70i}{17}
\end{aligned}
$$

Hence, rows are linearly independent.

As, $\mathbb{C}^3$ has 3 dimensions, and given vectors are linearly independent and span $\mathbb{C}$, they form a basis.

6.1.3 Review Metrics

- Time Spent:
- Confidence (1-5):
- Mistakes / Takeaways:

6.2 Block B: Theory & Proof Memorization

6.2.1 Task

6.2.2 Solution

6.2.3 Review Metrics

- Time Spent:

- Confidence (1-5):
- Mistakes / Takeaways:

6.3 Block C: Application/Testing

6.3.1 Task

6.3.2 Solution

6.3.3 Review Metrics

- Time Spent:
- Confidence (1-5):
- Mistakes / Takeaways:

7 Day 07: Synthesis - Maps & Fields

7.1 Block A: Active Computation

7.1.1 Task

1. verifying the definition of a field
   show that $\mho_2$ is indeed a field (see example ii.1.5 in the lecture notes).

7.1.2 Solution

1. (1)

   Definition (Field). A field $\mathbb{F}$ is a set equipped with additive and multiplicative operations ($+, \times$, respectively), such that it fullfils the following axioms:

   Let $a, b, c \in \mathbb{F}$.

   (a) Additive associativity
   $(a + b) + c = a + (b + c)$

   (b) Additive commutativity
   $a + b = b + a$

   (c) Additive identity
   $\exists 0 \in \mathbb{F} : \forall a \in \mathbb{F}, a + 0 = a$

(d) Additive inverse
$\forall a \in \mathbb{F}, \exists(-a) : a + (-a) = 0$

(e) Multiplicative associativity
$(a \times b) \times c = a \times (b \times c)$

(f) Multiplicative commutativity
$a \times b = b \times a$

(g) Multiplicative identity
$\exists 1 \in \mathbb{F} : 1 \neq 0, \forall a \in \mathbb{F}, a \times 1 = a$

(h) Multiplicative inverse
$\forall a \in \mathbb{F} : a \neq 0, \exists a^{-1} : a \times a^{-1} = 1$

(i) Distributivity
$a \times (b + c) = (a \times b) + (a \times c)$

7.1.3 Review Metrics

- Time Spent:
- Confidence (1-5):
- Mistakes / Takeaways:

7.2 Block B: Theory & Proof Memorization

7.2.1 Task

7.2.2 Solution

7.2.3 Review Metrics

- Time Spent:
- Confidence (1-5):
- Mistakes / Takeaways:

7.3 Block C: Application/Testing

7.3.1 Task

1. Provide the definition of a field. (4 Points)
2. Consider sets X and Y and a map $f : X \to Y$.

(a) Let $A \subseteq X$ be a subset. Define $f(A)$. (1 Point)
(b) Define the term injective for the map f. (1 Point)
(c) Is the map $f : \mathbb{R}_+ \to \mathbb{R}$ with $f(x) := x^4$ injective? ($\mathbb{R}_+$ is the set of all non-negative real numbers) (1 Point)
(d) Let Z be another set and $g : Y \to Z$ another map. Define the composition of f and g. (1 Point)
(e) Show that function composition is associative. (1 Point)

7.3.2 Solution

1. (1)

   Definition (Field). A field $\mathbb{F}$ is a set equipped with additive and multiplicative operations ($+, \times$, respectively), such that it fullfils the following axioms:

   Let $a, b, c \in \mathbb{F}$.

   (a) Additive associativity
   $(a + b) + c = a + (b + c)$
   (b) Additive commutativity
   $a + b = b + a$
   (c) Additive identity
   $\exists 0 \in \mathbb{F} : \forall a \in \mathbb{F}, a + 0 = a$
   (d) Additive inverse
   $\forall a \in \mathbb{F}, \exists(-a) : a + (-a) = 0$
   (e) Multiplicative associativity
   $(a \times b) \times c = a \times (b \times c)$
   (f) Multiplicative commutativity
   $a \times b = b \times a$
   (g) Multiplicative identity
   $\exists 1 \in \mathbb{F} : 1 \neq 0, \forall a \in \mathbb{F}, a \times 1 = a$
   (h) Multiplicative inverse
   $\forall a \in \mathbb{F} : a \neq 0, \exists a^{-1} : a \times a^{-1} = 1$
   (i) Distributivity
   $a \times (b + c) = (a \times b) + (a \times c)$

2. (2)

   (a) $f(A) := \{f(a) \mid a \in A\}$
   (b) f is injective $\iff \forall p, q \in X : f(p) = f(q) \implies p = q$
   (c) Yes, as every element of the given domain is mapped to exactly one element of codomain. Real positive number in power of 4 results in a real positive number.
   (d) $g \circ f : g(f(x)), \forall x \in X$

(e) $f : X \to Y$, $g : Y \to Z$, $h : Z \to W$
$(h \circ g) \circ f = h \circ (g \circ f)$
$(h \circ g) \circ f = (h \circ g)(f(x)) = h(g(f(x)))$
$h \circ (g \circ f) = h((g \circ f)(x)) = h(g(f(x)))$

7.3.3 Review Metrics

- Time Spent:
- Confidence (1-5):
- Mistakes / Takeaways:

8 Day 08: Synthesis & Pre-Mock Drilling

8.1 Block A: Active Computation

8.1.1 Task

1. (7 points) Determine, for the map

$$f : \mathbb{Z}_5^3 \to \mathbb{Z}_5^3, \quad x \mapsto \begin{pmatrix} 3x_1 + 4x_2 + 2x_3 \\ x_1 + 3x_2 + x_3 \\ 4x_1 + 2x_2 + 2x_3 \end{pmatrix}$$

the set of all points $x \in \mathbb{Z}_5^3$ with

$$f(x) = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}$$

8.1.2 Solution

$$\left[\begin{array}{ccc|c} 3 & 4 & 2 & 1 \\ 1 & 3 & 1 & 3 \\ 4 & 2 & 2 & 1 \end{array}\right]$$

$$
\begin{aligned}
1-(m_{21})(3) &\equiv 0 \quad (mod\, 5) \\
m_{21} &= 2 \\
(2,2) &= 3-(2)(4) = -5 \equiv 0 \\
(2,3) &= 1-(2)(2) = -3 \equiv 2 \\
(2,4) &= 3-(2)(1) = 1
\end{aligned}
$$

$$
\implies \left[\begin{array}{ccc|c} 3 & 4 & 2 & 1 \\ 0 & 0 & 2 & 1 \\ 4 & 2 & 2 & 1 \end{array}\right]
$$

$$
\begin{aligned}
4-(m_{31})(3) &\equiv 0 \quad (mod\, 5) \\
m_{31} &= 3 \\
(3,2) &= 2-(3)(4) = -10 \equiv 0 \\
(3,3) &= 2-(3)(2) = -4 \equiv 1 \\
(3,4) &= 1-(3)(1) = -2 \equiv 3
\end{aligned}
$$

$$
\implies \left[\begin{array}{ccc|c} 3 & 4 & 2 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & 3 \end{array}\right]
$$

Note that the second and third rows are dependent, as $R_3 \times 2 = R_2$.

$$
\implies \left[\begin{array}{ccc|c} 3 & 4 & 2 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]
$$

Hence,

$$
\begin{aligned}
(2)x_3 &\equiv 1 \quad (mod\, 5) && \implies x_3 = 3 \\
(0)x_2 + (0)x_3 &\equiv 0 \quad (mod\, 5) && \implies x_2 = n, \forall n \in \mathbb{Z}_5 \\
(3)x_1 + (4)x_2 + (2)x_3 &\equiv 1 \quad (mod\, 5) && \implies x_1 = 2n
\end{aligned}
$$

Thus, the solution set is:

$$\{(2n, n, 3) \mid \forall n \in \mathbb{Z}_5\}$$

8.1.3 Review Metrics

- Time Spent: hour, was confused with $\mathbb{Z}_5$.
- Confidence (1-5): 5
- Mistakes / Takeaways: none

8.2 Block B: Theory & Proof Memorization

8.2.1 Task

8.2.2 Solution

8.2.3 Review Metrics

- Time Spent:
- Confidence (1-5):
- Mistakes / Takeaways:

8.3 Block C: Application/Testing

8.3.1 Task

1. (2 points) Define the notions sum, direct sum, and complement for subspaces of a vector space.
2. (2 points) State the dimension theorem for sums of subspaces of a K -vector space.
3. (6 points) Prove the dimension theorem for sums of subspaces of a K -vector space. You do not have to prove additional results used in your proof, but you must clearly indicate which theorems/propositions you use (by stating them precisely).

8.3.2 Solution

1. (a)

   - $V + W := \{v + w : v \in V, w \in W\}$

- $V \oplus W$ is a sum where $V \cap W = \{0\}$
- W is a complement of V iff $V \oplus W = U$

2. (b) + (c)

   Let U_1, U_2 be finite subspaces of vector space V. Define $I := U_1 \cap U_2$ and choose basis $B_I = \{w_1, \cdots, w_k\}$. Extend B_I to basis U_1 and U_2:

$$B_{U_1} = \{w_1, \cdots, w_k, u_1, \cdots, u_p\} \quad B_{U_2} = \{w_1, \cdots, w_k, v_1, \cdots, v_q\}$$

   Claim: constructed $B = \{w_1, \cdots, w_k, u_1, \cdots, u_p, v_1, \cdots v_q\}$ is basis of $U_1 + U_2$.

   Span is immediate. Let $x \in U_1 + U_2$. It can be represented as $x = x_1 + x_2 : x_1 \in U_1, x_2 \in U_2$, hence it can be represented as scalar sum of their basis B_{U_1} and B_{U_2}.

   Linear independence. Assume $\sum \alpha_i w_i + \sum \beta_j u_j + \sum \gamma_l v_l = 0$. Rearange,

$$\sum \gamma_l v_l = -\left(\sum \alpha_i w_i + \sum \beta_j u_j\right) \in U_1$$

   As LHS is in U_2, the RHS must be in $U_1 \cap U_2$, and hence could be written as $\sum \mu_i w_i$. Thus,

$$\sum \gamma_l v_l - \sum \mu_i w_i = 0$$

   Linear independence of B_{U_2} forces all $\gamma_l = 0$. Substituting back:

$$\sum \alpha_i w_i + \sum \beta_j u_j = 0$$

   Linear independence of B_{U_1} forces $\alpha_i = \beta_j = 0$. Hence B is linearly independent.

   Therefore, as B is basis of $U_1 + U_2$,

$$dim(U_1 + U_2) = k + p + q = (p + k) + (q + k) - k = |U_1| + |U_2| - |U_1 \cap U_2|$$

8.3.3 Review Metrics

- Time Spent:
- Confidence (1-5):
- Mistakes / Takeaways:

9 Day 09: Timed Mock Exam 1

9.1 Block A: Active Computation (Warm-up)

9.1.1 Task

Let

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 6 & 3 & 4 \\ 3 & 1 & 5 \end{bmatrix} \in \mathbb{R}^{3\times 3}, \quad b^{(1)} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad b^{(2)} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

(a) Compute an LU factorization with pivoting, i.e. find P, L, U such that $PA = LU$.

(b) Using the same P, L, U, solve $Ax = b^{(1)}$ and $Ax = b^{(2)}$. Show the two solutions.

9.1.2 Solution

1. (a)

$$\begin{aligned} P_1 \left[A \mid b^{(1)} \right] &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \left[\begin{array}{ccc|c} 3 & 1 & 1 & 1 \\ 6 & 3 & 4 & 2 \\ 3 & 1 & 5 & 3 \end{array}\right] \\ &= \left[\begin{array}{ccc|c} 6 & 3 & 4 & 2 \\ 3 & 1 & 1 & 1 \\ 3 & 1 & 5 & 3 \end{array}\right] \end{aligned}$$

$$\begin{aligned} m_{21} &= \frac{3}{6} = \frac{1}{2} \\ (2,2) &= 1 - (\frac{1}{2})(3) = \frac{2}{2} - \frac{3}{2} = -\frac{1}{2} \\ (2,3) &= 1 - (\frac{1}{2})(4) = 1 - 2 = -1 \\ (2,4) &= 1 - (\frac{1}{2})(2) = 0 \end{aligned}$$

$$\Longrightarrow \left[\begin{array}{ccc|c} 6 & 3 & 4 & 2 \\ 0 & -\frac{1}{2} & -1 & 0 \\ 3 & 1 & 5 & 3 \end{array}\right]$$

$$
\begin{aligned}
m_{31} &= \frac{1}{2} \\
(3,2) &= 1 - (\frac{1}{2})(3) = \frac{2}{2} - \frac{3}{2} = -\frac{1}{2} \\
(3,3) &= 5 - (\frac{1}{2})(4) = 5 - 2 = 3 \\
(3,4) &= 3 - (\frac{1}{2})(2) = 3 - 1 = 2
\end{aligned}
$$

$$
\Longrightarrow \left[\begin{array}{ccc|c} 6 & 3 & 4 & 2 \\ 0 & -\frac{1}{2} & -1 & 0 \\ 0 & -\frac{1}{2} & 3 & 2 \end{array}\right]
$$

$$
\begin{aligned}
m_{32} &= 1 \\
(3,3) &= 3 - (1)(-1) = 4 \\
(3,4) &= 2 - (1)(0) = 2
\end{aligned}
$$

$$
\Longrightarrow \left[\begin{array}{ccc|c} 6 & 3 & 4 & 2 \\ 0 & -\frac{1}{2} & -1 & 0 \\ 0 & 0 & 4 & 2 \end{array}\right]
$$

Hence,

$$
\begin{aligned}
P &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
L &= \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{1}{2} & 1 & 1 \end{bmatrix} \\
U &= \begin{bmatrix} 6 & 3 & 4 \\ 0 & -\frac{1}{2} & -1 \\ 0 & 0 & 4 \end{bmatrix}
\end{aligned}
$$

2. (b)

- $b^{(1)}$

$$
\left[\begin{array}{ccc|c} 6 & 3 & 4 & 2 \\ 0 & -\frac{1}{2} & -1 & 0 \\ 0 & 0 & 4 & 2 \end{array}\right]
$$

$$
\begin{aligned}
(4)x_3 &= 2 && \implies x_3 = \frac{1}{2} \\
(-\frac{1}{2})x_2 + (-1)x_3 &= 0 && \implies x_2 = -1 \\
(6)x_1 + (3)x_2 + (4)x_3 &= 2 && \implies x_1 = \frac{1}{2}
\end{aligned}
$$

9.1.3 Review Metrics

- Time Spent:
- Confidence (1-5):
- Mistakes / Takeaways:

9.2 Block B: Theory & Proof Memorization (Refresher)

9.2.1 Task

9.2.2 Solution

9.2.3 Review Metrics

- Time Spent:
- Confidence (1-5):
- Mistakes / Takeaways:

9.3 Block C: Application/Testing (Full Mock Exam)

9.3.1 Task

9.3.2 Solution

9.3.3 Review Metrics

- Time Spent:
- Total Exam Score Estimate:
- Time Management Assessment:
- Mistakes / Takeaways:

10 Day 10: Timed Mock Exam 2

10.1 Block A: Active Computation (Warm-up)

10.1.1 Task

10.1.2 Solution

10.1.3 Review Metrics

- Time Spent:
- Confidence (1-5):
- Mistakes / Takeaways:

10.2 Block B: Theory & Proof Memorization (Refresher)

10.2.1 Task

10.2.2 Solution

10.2.3 Review Metrics

- Time Spent:
- Confidence (1-5):
- Mistakes / Takeaways:

10.3 Block C: Application/Testing (Full Mock Exam)

10.3.1 Task

10.3.2 Solution

10.3.3 Review Metrics

- Time Spent:
- Total Exam Score Estimate:
- Time Management Assessment:
- Mistakes / Takeaways:

11 Day 11: Buffer, Review & Rest

11.1 Block A: Active Computation (Review)

11.1.1 Task

11.1.2 Solution

11.1.3 Review Metrics

- Time Spent:
- Confidence (1-5):
- Mistakes / Takeaways:

11.2 Block B: Theory & Proof Memorization (Review)

11.2.1 Task

11.2.2 Solution

11.2.3 Review Metrics

- Time Spent:
- Confidence (1-5):
- Mistakes / Takeaways:

11.3 Block C: Application/Testing (Analysis & Consolidation)

11.3.1 Task

Review all notes, mistakes, and confidence levels from Days 1-10.

11.3.2 Solution

- Top 3 recurring calculation errors:
- Weakest proof(s) to read one last time:
- Final Strategy / Exam Execution Plan:

12 Extra

12.1 P1

- Definition of a vector space
  A vector space V over a field $\mathbb{F}$, has two operations:
  - Additive: $+ : V + V \to V$
  - Multiplicative: $\times : \mathbb{F} \times V \to V$

And must satisfy the following axioms:

1. Additive associativity
2. Additive commutativity
3. Additive identity
4. Additive inverse
5. Distributivity over scalar addition
6. Distributivity over vector addition
7. Compatability of scalar multiplication
8. Scalar multiplicative identity

12.2 P2

- Definition of preimage
  Let $f : U \to V$, and $S \subseteq V$.
  $f^{-1}(S) = \{u \in U : f(u) = S\}$
- Definition of surjectivity
  Let $f : U \to V$.
  $\forall v \in V, \exists u \in U : f(u) = v$
- Surjectivity proof
  Proof $f : X \to Y$ is surjective.
  1. Let $y \in Y$ be arbitrary.
  2. Choose $x \in X$ such that $f(x) = y$ (creative and needs scratch work).
  3. Verify that $x \in X$.
  4. Verify that $f(x) = y$.

5. Conclude surjection.

- Definition of inverse map
  Let $f : X \to Y$.
  $f^{-1}(f(x)) = x, \forall x \in X$ and $f(f^{-1}(y)) = y, \forall y \in Y$.
  Where f^{-1} is inverse map.

12.3 P3

- LU factorisation

12.4 P4

- Inverse matrix

12.5 P5

- Definition of sum
  $U + V = \{u + v : u \in U, v \in V\}$
- Definition of direct sum
  $U \oplus V$ is $U + V$ wich satisfies $U \cap V = \{0\}$
- Definition of complementary
  Subspace U is complementary of subspace V, if $U \oplus V = W$, where W is a vector space.
- Statement and proof of Sum of Subspaces Theorem.
  Known.

12.6 P6